

A Level Mechanics

Notes · Edexcel A Level Mathematics — Mechanics · Chapter 2: Kinematics — Constant Acceleration

Chapter 2: Kinematics — Constant Acceleration

Why this chapter matters

Constant acceleration kinematics is the foundation on which every later mechanics chapter is built: forces (Chapter 3) are analysed through the accelerations they cause, projectiles (Chapter 7) are just two independent constant-acceleration problems running at once, and variable acceleration (Chapter 4) is what you get when you replace the constant- a equations here with calculus. Getting comfortable with three habits now will pay off throughout the course: using explicit vector notation, stating a sign convention at the start of every solution, and being fluent enough with all five suvat equations to choose the most efficient one for a given problem, rather than defaulting to the same one or two every time.

2.1 Displacement, velocity, speed and acceleration

Concept

- **Displacement (s)** is a vector: the straight-line distance from a starting point, together with a direction.
- **Distance** is the corresponding scalar: the total length of the path travelled. It is always positive and never “cancels out”, even if the direction changes.
- **Velocity (v)** is the vector rate of change of displacement with time.
- **Speed** is the scalar magnitude of velocity.
- **Acceleration (a)** is the vector rate of change of velocity with time. A negative acceleration in the chosen positive direction is often called a **deceleration** or **retardation** — but the underlying vector quantity is still just “acceleration”. Mark schemes want a signed value, not a separate word for negative acceleration.

Because every one of these quantities (except distance and speed) is a vector, every one-dimensional problem must begin by fixing a positive direction. A numerical answer to a vector part of a question is incomplete — and loses marks — without a direction attached to it.

Key vocabulary

Term	Meaning
Displacement (s)	Vector: signed distance from a fixed starting point
Distance	Scalar: total path length travelled, always non-negative
Velocity (v)	Vector: rate of change of displacement with time
Speed	Scalar: magnitude of velocity

Acceleration (a)	Vector: rate of change of velocity with time
Deceleration / retardation	Informal name for a negative acceleration in the chosen positive direction — not a separate signed quantity
Initial velocity (u), final velocity (v)	The standard suvat letters used throughout this course; mark schemes reference them by name, so get comfortable using u and v specifically rather than substituting your own labels

Key formulas

KEY FORMULA

$$v = ds/dt, \quad a = dv/dt$$

For now (constant acceleration only) these can be treated as “velocity is the gradient of a displacement–time graph” and “acceleration is the gradient of a velocity–time graph” (Section 2.3); the derivative forms above are properly needed once acceleration is no longer constant (Chapter 4).

Worked example

EXAMPLE 2.1

Question. A cyclist rides 120 m due east, then turns around and rides 50 m due west, all in a total time of 40 s. Find (a) the total distance travelled, (b) the displacement from the start, (c) the average speed, (d) the average velocity.

Step 1 (part a) — total distance. Distance is the total path length, regardless of direction, so add the two legs of the journey together:

$$\text{distance} = 120 + 50 = 170 \text{ m} \quad \text{M1 A1}$$

Step 2 (part b) — displacement. Displacement is the signed straight-line distance from the start. Taking east as positive, the return leg is –50 m:

$$s = 120 - 50 = 70 \text{ m (east)} \quad \text{M1 A1 (direction required for the A mark)}$$

Step 3 (part c) — average speed. Average speed uses total distance, not displacement:

$$\text{average speed} = \text{total distance} / \text{total time} = 170/40 = 4.25 \text{ m s}^{-1} \quad \text{M1 A1}$$

Step 4 (part d) — average velocity. Average velocity uses displacement, not distance — this is exactly why the two answers differ:

$$\text{average velocity} = s/t = 70/40 = 1.75 \text{ m s}^{-1} \text{ (east)} \quad \text{M1 A1}$$

Answer: (a) 170 m, (b) 70 m east, (c) 4.25 m s^{–1}, (d) 1.75 m s^{–1} east.

Exam tip

EXAM TIP

It is easy to treat distance/speed and displacement/velocity as interchangeable, and the difference only becomes visible once the motion changes direction — exactly as it does in

Example 2.1, where the cyclist turns around. Build the habit early: write “taking [direction] as positive” as the very first line of every kinematics solution, before any numbers are substituted. Once that convention is fixed, deciding whether an answer needs a direction (a vector) or not (a scalar) becomes mechanical rather than a source of lost marks.

Common exam mistakes

- Not fixing a positive direction before assigning signed values, leading to inconsistent signs partway through a solution.
- Giving a magnitude only for a vector answer, with no direction stated.
- Treating distance and displacement (or speed and velocity) as interchangeable when the motion changes direction.

2.2 The suvat equations

Concept

For motion in a straight line with **constant acceleration**, five variables are linked: initial velocity u , final velocity v , acceleration a , displacement s , and time t . Because acceleration is constant, these five variables are connected by five equations — collectively called the **suvat equations** — each of which involves only four of the five variables. Solving a suvat problem is simply a matter of identifying the three variables that are known, the one variable that is wanted, and then picking the single equation that omits the fifth (unwanted) variable.

Key vocabulary

Term	Meaning
suvat	The five variables s , u , v , a , t and, informally, the set of equations linking them
Uniform acceleration	Acceleration that is constant in both magnitude and direction — required for the suvat equations to apply
Knowns / unknowns	The variables given in a question versus the one(s) being found — identifying these correctly is the main problem-solving skill of this section

Key formulas (derivations)

KEY FORMULAS

Starting from the definition of constant acceleration as the (constant) rate of change of velocity:

$$a = (v - u)/t \implies v = u + at \quad (1)$$

Since acceleration is constant, velocity increases uniformly, so the average velocity over the interval is the mean of the initial and final velocities:

$$s = ((u + v)/2) t \quad (2)$$

Substituting (1) into (2) to eliminate v :

$$s = ((u + (u + at))/2) t = ut + \frac{1}{2}at^2 \quad (3)$$

Substituting (1) into (2) to eliminate u instead (using $u = v - at$):

$$s = vt - \frac{1}{2}at^2 \quad (4)$$

Eliminating t between (1) and (2): from (1), $t = (v - u)/a$; substituting into (2):

$$s = ((u + v)/2)((v - u)/a) = (v^2 - u^2)/(2a) \implies v^2 = u^2 + 2as \quad (5)$$

Equation	Variable not involved
$v = u + at$	s
$s = ((u+v)/2)t$	a
$s = ut + \frac{1}{2}at^2$	v
$s = vt - \frac{1}{2}at^2$	u
$v^2 = u^2 + 2as$	t

Procedure

1. List the five suvat variables and write down the three that are given, using a consistent sign convention (state the positive direction).
2. Identify the variable the question wants, and the one variable that is neither given nor wanted.
3. Select the single equation from the table above that omits that unwanted variable, and substitute directly — never solve for an unneeded intermediate variable first.
4. Check the sign and units of the answer against the physical situation (e.g. a deceleration should come out negative if the positive direction was chosen as the direction of initial motion).

Worked example

EXAMPLE 2.2

Question. A car accelerates uniformly from rest and reaches a speed of 24 m s^{-1} after 6 s. Find (a) its acceleration, (b) the distance it travels in this time.

Step 1 — list the knowns. Taking the direction of motion as positive: $u = 0$, $v = 24$, $t = 6$. **B1**

Step 2 (part a) — find a. We want a, and s is not involved, so use $v = u + at$:

$$24 = 0 + 6a \quad \text{M1}$$

$$a = 24/6 = 4 \text{ m s}^{-2} \quad \text{A1}$$

Step 3 (part b) — find s. We want s, and a is now known, so any of the s-equations works. Use $s = ((u+v)/2)t$, which avoids re-using the rounded value of a:

$$s = ((0 + 24)/2)(6) \quad \text{M1}$$

$$s = 12 \times 6 = 72 \text{ m} \quad \text{A1}$$

Answer: (a) $a = 4 \text{ m s}^{-2}$, (b) $s = 72 \text{ m}$.

Exam tip

EXAM TIP

When several suvat equations could technically be used, it is tempting to reach for whichever one you remember best — usually $v = u + at$ or $s = ut + \frac{1}{2}at^2$ — rather than the one that avoids the unwanted variable. This habit causes two problems: it can add an unnecessary extra step, and, worse, it can pull a rounded intermediate value (like the $a = 4$ above) into a later calculation, introducing a rounding error that costs an accuracy mark even though the method is correct. Practise the variable-identification step (Procedure, steps 1–2) as something you do before writing any equation, not after.

Common exam mistakes

- Substituting into an equation that still contains an unknown fifth variable.
- Using a rounded intermediate answer (e.g. a rounded acceleration) in a later step instead of the exact value or the original data.
- Sign errors: forgetting that a deceleration is negative in the chosen positive direction, so $v^2 = u^2 + 2as$ needs a negative a .
- Mixing units (e.g. a speed given in km/h) without converting to SI (m s^{-1}) first.

2.3 Displacement–time and velocity–time graphs

Concept

Displacement–time (s–t) graph plots position against time.

- Its gradient at any point gives the velocity at that instant.
- A straight line means constant velocity; a curve means changing velocity (non-uniform motion); a horizontal section means the particle is at rest.

Velocity–time (v–t) graph plots velocity against time.

- Its gradient gives acceleration.
- Crucially, the area between the graph and the time axis gives displacement.
- A straight line on a v–t graph means constant (uniform) acceleration — exactly the suvat situation of Section 2.2. A curve means the acceleration is changing (covered in Chapter 4).

Area below the time axis represents negative (backward) displacement and must be subtracted, not added, when total displacement — rather than total distance — is required.

Key vocabulary

Term	Meaning
Gradient of s–t graph	Velocity at that instant
Gradient of v–t graph	Acceleration at that instant

Area under v–t graph	Displacement over that time interval (signed: area below the axis is negative)
Stationary	$v = 0$ — a horizontal section on an s–t graph, or a point where a v–t graph crosses the time axis
Non-uniform motion	Changing acceleration — shown as a curve on a v–t graph

Key formulas

KEY FORMULA

$v = \text{gradient of the s–t graph}$, $a = \text{gradient of the v–t graph}$, $s = \text{area under the v–t graph}$

For a trapezium-shaped region (a common shape under a multi-stage v–t graph) with parallel sides a and b and height (perpendicular width) h :

$$s = \frac{1}{2}(a + b)h$$

Worked example

EXAMPLE 2.3

Question. A particle starts from rest and accelerates uniformly to a speed of 15 m s^{-1} in 5 s. It then travels at this constant speed for a further 10 s, before decelerating uniformly to rest in 3 s. (a) Sketch the velocity–time graph. (b) Find the acceleration in each of the three stages. (c) Find the total distance travelled.

Step 1 (part a) — sketch the graph. Plot velocity (m s^{-1}) against time (s), with three straight-line stages:

- Stage 1: a straight line from (0, 0) to (5, 15) — the acceleration phase.
- Stage 2: a horizontal line from (5, 15) to (15, 15) — the constant-speed phase.
- Stage 3: a straight line from (15, 15) to (18, 0) — the deceleration phase.

Both axes must be labelled with quantity and units. **B1**

Step 2 (part b) — find the accelerations. Acceleration in each stage is the gradient of that stage:

$$a_1 = (15 - 0)/5 = 3 \text{ m s}^{-2}, \quad a_2 = (15 - 15)/10 = 0, \quad a_3 = (0 - 15)/3 = -5 \text{ m s}^{-2} \quad \text{M1 A1 (all three)}$$

Step 3 (part c) — find the total distance. Total distance is the total area under the graph. Split the region into a triangle (stage 1), a rectangle (stage 2) and a triangle (stage 3), then add the three areas:

$$s = (\frac{1}{2} \times 5 \times 15)_{\text{stage 1}} + (10 \times 15)_{\text{stage 2}} + (\frac{1}{2} \times 3 \times 15)_{\text{stage 3}} \quad \text{M1}$$

$$s = 37.5 + 150 + 22.5 = 210 \text{ m} \quad \text{A1}$$

Answer: (b) 3 m s^{-2} , 0, -5 m s^{-2} ; (c) 210 m.

Exam tip

EXAM TIP

This “sketch it, then use the graph to find a quantity” style of multi-stage question rewards two things separately: a correctly labelled sketch (missing units or axis labels loses the B1 mark on its own) and a total area correctly decomposed into simple shapes — usually triangles, rectangles and trapezia. It is worth stating explicitly that once velocity can be negative (motion reversing), area below the axis subtracts for displacement but still adds for total distance — this distinction causes far more lost marks than the graph-reading itself.

Common exam mistakes

- Confusing gradient (velocity or acceleration) with area (displacement) — reading the wrong feature of the graph.
- Treating a curved section as if it had constant acceleration, or a straight section as if acceleration were changing.
- Adding area below the time axis instead of subtracting it when displacement (not distance) is required.
- Leaving a sketch unlabelled, with no scale, units or axis names.

2.4 Vertical motion under gravity

Concept

An object moving vertically, close to the Earth's surface and with air resistance ignored, has **constant acceleration $g = 9.8 \text{ m s}^{-2}$ directed downwards** — so every vertical motion problem is simply a suvat problem (Section 2.2) once a sign convention is fixed. Three situations come up repeatedly:

- **Dropped:** $u = 0$.
- **Projected upwards:** decelerates under gravity, reaches $v = 0$ at its highest point, then falls back.
- **Projected downwards:** accelerates throughout.

The sign convention must be chosen and stated before substituting: if upwards is taken as positive, then $a = -g$ and any downward velocity or displacement is negative; if downwards is taken as positive, then $a = +g$ and any upward quantity is negative.

Key vocabulary

Term	Meaning
g	Acceleration due to gravity, taken as 9.8 m s^{-2} unless a question states otherwise
Free fall	Motion under gravity alone, air resistance modelled as negligible
Time of flight	Total time from projection until the object returns to its starting height (or hits the ground)
Maximum height	The point where the object is momentarily at rest, $v = 0$ (not $a = 0$ —

	acceleration is g throughout)
Projected upwards / released from rest / projected downwards	Standard phrasing for the initial condition, fixing the sign of u

Key formulas

KEY FORMULA

The same five suvat equations as Section 2.2, with $a = g$ (magnitude 9.8 m s^{-2}) substituted according to the chosen sign convention, e.g. taking upwards as positive:

$$v = u - gt, \quad s = ut - \frac{1}{2}gt^2, \quad v^2 = u^2 - 2gs$$

Worked example

EXAMPLE 2.4

Question. A ball is thrown vertically upwards from ground level with an initial speed of 14 m s^{-1} . Taking upwards as positive and $g = 9.8 \text{ m s}^{-2}$, find (a) the maximum height reached, (b) the total time for the ball to return to the ground.

Step 1 — state the model and sign convention. The ball is modelled as a particle moving freely under gravity alone (air resistance ignored). Taking upwards as positive: $u = 14$, $a = -9.8$.

B1

Step 2 (part a) — find the maximum height. At maximum height, $v = 0$. We want s and t is not involved, so use $v^2 = u^2 + 2as$:

$$0 = 14^2 + 2(-9.8)s \quad \text{M1}$$

$$s = 196/19.6 = 10 \text{ m} \quad \text{A1}$$

Step 3 (part b) — find the total time of flight. First find the time to reach maximum height, using $v = u + at$ with $v = 0$:

$$0 = 14 - 9.8t \quad \text{M1}$$

$$t = 14/9.8 = 10/7 \approx 1.43 \text{ s}$$

By symmetry of the up-and-down motion (the ball returns to the same height with equal speed), the total time of flight is twice this: **M1**

$$T = 2 \times 10/7 \approx 2.86 \text{ s} \quad \text{A1}$$

Answer: (a) 10 m, (b) approximately 2.86 s.

Exam tip

EXAM TIP

Vertical motion problems can feel more procedural than conceptual once the underlying physics is familiar — the real skill being tested is discipline in bookkeeping. Mark schemes expect the sign convention and modelling assumption (particle, air resistance negligible — linking back to Chapter 1) to be stated explicitly as part of the solution, not left implicit, and expect fluent

switching between whichever convention best suits a question's phrasing, rather than always defaulting to the same one. This section is also the direct setup for Chapter 7 (Projectiles), where this same vertical suvat analysis runs alongside an independent, constant-velocity horizontal one.

Common exam mistakes

- Switching sign convention partway through a solution (e.g. using $+9.8$ after declaring upwards as positive).
- Treating maximum height as $a = 0$ rather than $v = 0$ (acceleration is g throughout the flight).
- Omitting the modelling assumption that air resistance is negligible.
- Accepting a negative time, or a spurious second root, as a final answer without rejecting it on physical grounds.

Chapter summary

Idea	Key point
Displacement/velocity vs distance/speed	Vectors need a direction stated; scalars do not
suvat equations	Five equations linking u , v , a , s , t — each omits exactly one variable; pick the one that avoids the unknown you don't need
s–t graph	Gradient = velocity
v–t graph	Gradient = acceleration; area between graph and time axis = displacement (signed)
Vertical motion under gravity	A suvat problem with $a = \pm g$; fix and state the sign convention before substituting

You should now be able to

- Distinguish displacement, velocity and acceleration (vectors) from distance and speed (scalars), stating a direction with every vector answer.
- Derive and select the correct suvat equation for a given set of knowns and unknowns, without introducing unnecessary intermediate steps.
- Sketch and interpret displacement–time and velocity–time graphs, including multi-stage journeys, and find distance or displacement from the area under a v – t graph.
- Solve problems on vertical motion under gravity using $g = 9.8 \text{ m s}^{-2}$, with a clearly stated and consistently applied sign convention, including objects dropped, thrown upwards or projected downwards.